先进计算与数字工程研究中心

PATTERN RECOGNITION AND MACHINE LEARNING

Lecture 08 – Linear Models for Regression and Classification

Chizhi Chris ZHANG

zhangchizhi@ciomp.ac.cn

Advanced Computing and Digital Technology Research Center

University of Chinese Academy of Sciences

Fall 2026

先进计算与数字工程研究中心
ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

One customer, three different outputs

Regression

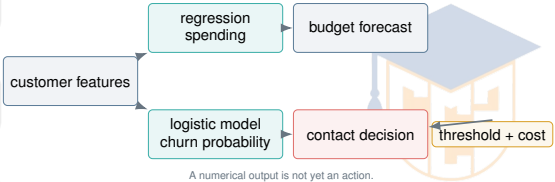
Predict next month's spending as a number.

Probability

Estimate the probability of subscription cancellation.

Decision

A probability becomes an action only after a threshold and a cost are chosen.



ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Five syllabus blocks in three hours

Syllabus block	Time	Main question
Linear regression	1.0 h	How is a number fitted?
Regression performance metrics	0.5 h	Which residuals matter?
Logistic regression	0.5 h	How is probability modelled?
Classification performance metrics	0.5 h	Which decisions count as good?
Imbalanced classification	0.5 h	What changes when positives are rare?



先进计算与数字工程研究中心
ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

The lecture follows prediction into action

1. Encode observations as a design matrix.
2. Fit a numerical response with least squares.
3. Audit residuals with several regression metrics.
4. Map a linear score to probability with a sigmoid.
5. Choose classification metrics and thresholds for the real cost.

A rent prediction starts with a scatterplot

Core idea

Inspect the response, candidate predictors, ranges, missingness, and unusual points before fitting.

Concrete case

Apartment rent may depend on area, distance to a station, floor, and neighborhood.

Watch for

A fitted line cannot repair a measurement unit error or a biased sample.



Plots reveal direction, spread, and possible nonlinearity.

先进计算与数字工程研究中心
ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

The design matrix turns many equations into one

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

Object	Shape	Meaning
$\mathbf{X}$	$N \times (p+1)$	Rows are observations
$\boldsymbol{\beta}$	$(p+1) \times 1$	Unknown coefficients
$\mathbf{y}$	$N \times 1$	Observed responses
$\boldsymbol{\varepsilon}$	$N \times 1$	Unexplained variation

The intercept is the model's reference level

What the model says

The intercept β_0 is the prediction when all encoded predictors equal zero.

What it does not say

Without meaningful coding, the intercept may be mathematically valid but practically uninteresting.

Use it here

After centering floor area, the intercept becomes the predicted rent for an average sized apartment.

A residual is an observed prediction error

Core idea

For observation i, the residual is $e_i = y_i - \hat{y}_i$.

Concrete case

A rent of 4200 with a prediction of 3900 gives a residual of 300.

Watch for

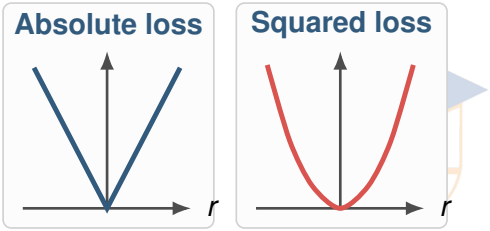
Residuals are sample quantities; they are not the unobservable population noise itself.

Squaring residuals changes the meaning of error

Residual	Absolute contribution	Squared contribution
1	1	1
2	2	4
5	5	25
10	10	100

Consequence

Large misses dominate the squared error objective.



Squaring gives large residuals disproportionate influence.

Ordinary least squares chooses the smallest sum of squares

$$\widehat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \sum_{i=1}^{N} (y_i - \mathbf{x}_i^\top \boldsymbol{\beta})^2$$

Equivalent form

Minimize $\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2$.

Interpretation

Choose the coefficient vector whose predictions lie closest to the observed responses in squared distance.

Least squares is an orthogonal projection

What the model says

The fitted vector $\widehat{\mathbf{y}}$ is the projection of $\mathbf{y}$ onto the column space of $\mathbf{X}$.

What it does not say

Projection describes fitted responses, not necessarily the true data generating mechanism.

Use it here

The residual vector is orthogonal to every column of $\mathbf{X}$ at the optimum.

The gradient reveals the normal equations

$$\nabla_{\boldsymbol{\beta}} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 = -2\mathbf{X}^\top(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$$

Set to zero

$\mathbf{X}^\top \mathbf{X}\widehat{\boldsymbol{\beta}} = \mathbf{X}^\top \mathbf{y}$.

Meaning

Zero gradient is sufficient here because the objective is convex.

The closed form solution needs an invertible Gram matrix

$$\widehat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

Condition

The inverse exists only when the columns of $\mathbf{X}$ are linearly independent.

Computation

Numerical software usually solves a linear system or uses a factorization instead of forming the inverse.

Full column rank gives a unique coefficient vector

- If $\operatorname{rank}(\mathbf{X}) = p + 1$, the least squares coefficient vector is unique.
- A duplicated temperature column creates infinitely many coefficient pairs with the same fitted values.

Decision check

Unique predictions can exist even when individual coefficients are not uniquely identified.

Singular values expose weak data directions

Core idea

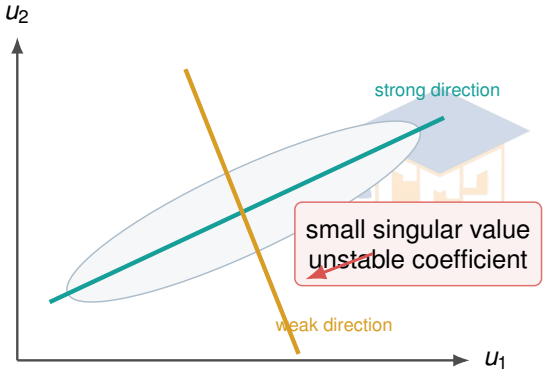
Small singular values identify directions in coefficient space that the data barely constrain.

Concrete case

Two nearly duplicate columns create a direction where one coefficient rises while the other falls.

Watch for

A tiny singular value amplifies measurement noise in the estimated coefficients.



The pseudoinverse selects one least squares solution

$$\widehat{\boldsymbol{\beta}} = \mathbf{X}^{+}\mathbf{y}$$

Role

The Moore–Penrose pseudoinverse works when the design matrix is rank deficient.

Convention

Among exact least squares solutions, it returns the minimum Euclidean norm solution.

Closed form and gradient descent solve the same objective differently

Aspect	Factorization or solve	Gradient descent
Small dense problem	Usually efficient	Often unnecessary
Very large problem	May be expensive	Scales by iterations
Feature scaling	Still useful	Often essential
Approximation	Near machine precision	Controlled by stopping



A few lines implement gradient descent

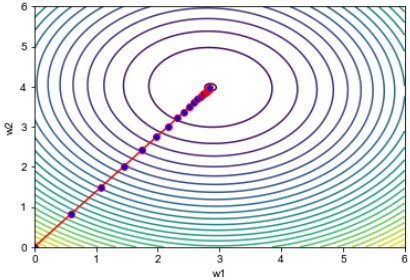
先进计算与数字工程研究中心
ADVANCED COMPUTING AND DIGITAL ENGINEERING RESEARCH CENTER

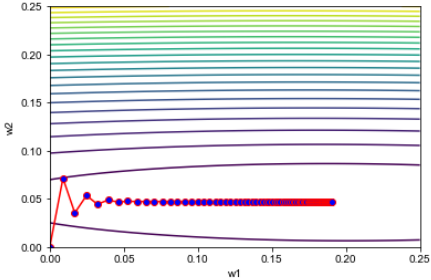
```
import numpy as np
X = np.c_[np.ones(5), np.arange(5.0)]
y = np.array([1.0, 2.1, 2.9, 4.2, 5.1])
beta = np.zeros(2)
for _ in range(2000):
    gradient = 2 * X.T @ (X @ beta - y) / len(y)
    beta -= 0.05 * gradient
print(np.round(beta, 3))
```

Read the result

The first value is the intercept; the second is the slope.

The learning rate controls step size, not model capacity



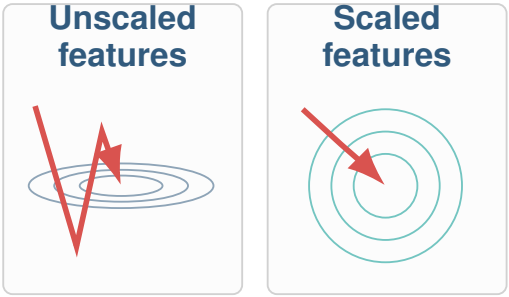


Core idea. A small learning rate moves cautiously; a large one may overshoot the minimum.

Concrete case. Walking down a narrow valley requires shorter steps across the steep direction.

Watch for. Changing the learning rate does not add or remove predictors from the hypothesis class.

Feature scaling reshapes the optimization landscape



Scaling improves conditioning, not model capacity.

Start from the case

Area measured in square meters and distance measured in kilometers otherwise create very different gradient magnitudes.

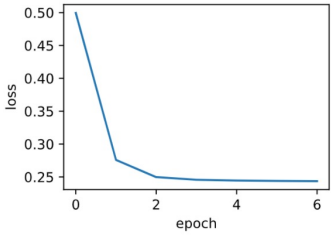
Model statement

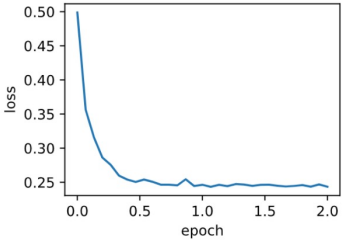
Standardization puts predictors on comparable numerical scales.

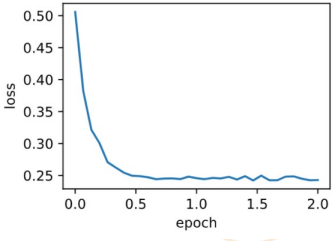
Common mistake

Scaling changes coefficient units; it does not erase the original meaning if the transformation is recorded.

Convergence curves separate slow learning from divergence







- Plot objective value against iteration and verify that it decreases toward a stable level.
- A zigzagging curve suggests an excessive learning rate; a flat high curve suggests tiny steps or poor scaling.

Decision check. A decreasing training objective alone does not prove good validation performance.

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Polynomial regression is still a linear model

What the model says

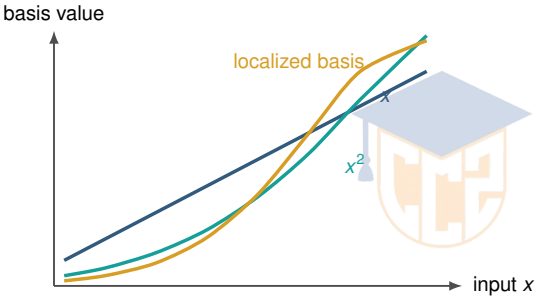
Create columns $x, x^2, \ldots, x^d$ and fit their coefficients linearly.

What it does not say

High degree features can interpolate noise and extrapolate wildly.

Use it here

A curved demand pattern can be represented by a quadratic basis.



The model is linear in coefficients, not necessarily in raw input.

Basis functions turn domain knowledge into columns

Core idea

Write $\hat{y} = \beta_0 + \sum_{j=1}^{m} \beta_j \phi_j(x)$ with fixed basis functions ϕ_j.

Concrete case

Hourly electricity demand may use periodic sine and cosine bases.

Watch for

A basis chosen after seeing test results leaks evaluation information.

Categorical variables need explicit coding

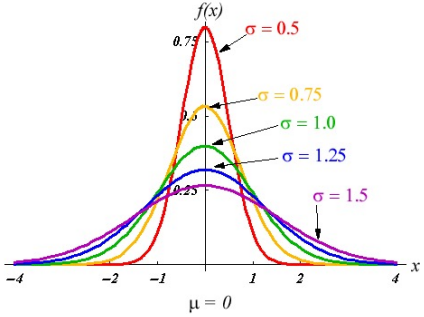
- Use indicator columns and choose a reference category when an intercept is present.
- Neighborhood A can be the reference, with separate differences for B and C.



Decision check

Including every category indicator plus an intercept creates exact collinearity.

Gaussian noise connects least squares to likelihood



What the model says

Assume $y_i \mid \mathbf{x}_i \sim \mathcal{N}(\mathbf{x}_i^\top \boldsymbol{\beta}, \sigma^2)$.

What it does not say

The assumption concerns conditional errors, not whether every predictor is normally distributed.

Use it here

Repeated demand observations fluctuate symmetrically around the model mean.

Maximum likelihood recovers least squares

$$-\log p(\mathbf{y} \mid \mathbf{X}, \beta) = \frac{1}{2\sigma^2} \sum_i (y_i - \mathbf{x}_i^\top \beta)^2 + C$$

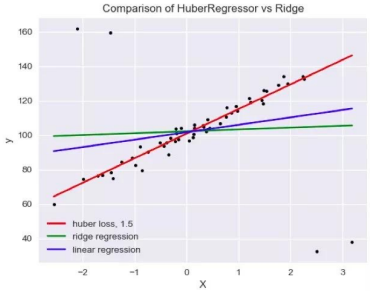
Conclusion

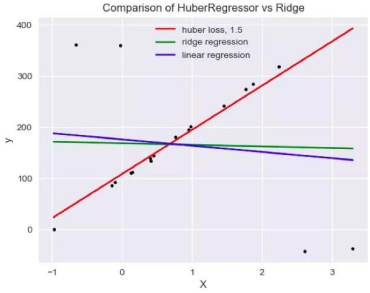
For fixed variance, maximizing Gaussian likelihood is equivalent to minimizing squared residuals.

Boundary

The equivalence inherits the modelling assumptions.

Outliers can pull a least squares line





Start from the case. One wrongly recorded house price can rotate the fitted line.

Model statement. Squared loss gives a point with a residual of ten one hundred times the weight of a residual of one.

Common mistake. Deleting an unusual point only because it hurts the fit is not a valid cleaning rule.

Residual plots test the story left after fitting

What the model says

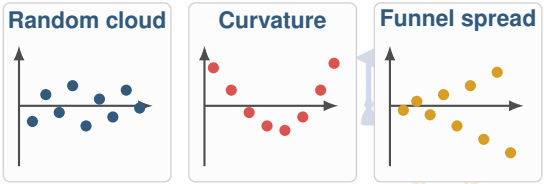
Plot residuals against fitted values and important predictors; look for curvature, changing spread, and clusters.

What it does not say

A random looking residual plot does not prove every assumption, but visible structure reveals a missed pattern.

Use it here

A funnel shape suggests error variance grows with predicted sales.



Residual structure is evidence that the model story is incomplete.

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

A regression coefficient describes a conditional association

- In $\widehat{y} = b + \sum_j w_j x_j$, w_j is the predicted change per unit of x_j.
- The comparison holds other included columns fixed.
- Units and feature coding determine the interpretation.
- Observational regression alone does not establish a causal effect.

Coefficient uncertainty asks how the fit would vary

Start from the case

A stable total prediction can coexist with unstable individual advertising channel coefficients.

Model statement

Standard errors or resampling summarize coefficient variation across hypothetical repeated samples.

Common mistake

A narrow interval under a wrong sampling model is not trustworthy evidence.

Advertising data connect coefficients to a real decision

- Fit sales from multiple advertising channels, then compare held out predictions and coefficient stability.
- A channel with a large coefficient may still have little practical impact if its feasible budget range is tiny.

Decision check

Coefficient magnitude is not comparable across predictors with different units.



Association strength differs by channel.

先进计算与数字工程研究中心
ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Advertising model performance

What the model says

Show predictions, residuals, and a held out metric in the same review.

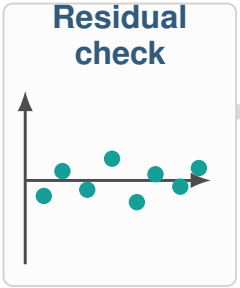
What it does not say

A single average metric can hide systematic failure at the edge.

Use it here

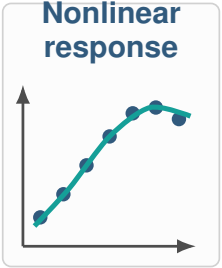
An advertising model may track the central range well but miss high spending campaigns.

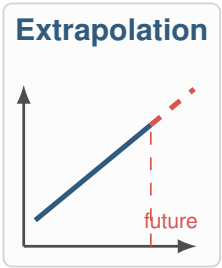




Fit and leftover structure answer different questions.

A bicycle demand case tests extrapolation





A straight extrapolation can leave the observed support.

Core idea

Time, weather, season, and calendar features build a transparent demand baseline.

Concrete case

A model trained in spring may fail during an unseen holiday heat wave.

Watch for

Good interpolation within the training range does not guarantee extrapolation outside it.

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Regularization stabilizes an overexpanded linear model

Start from the case

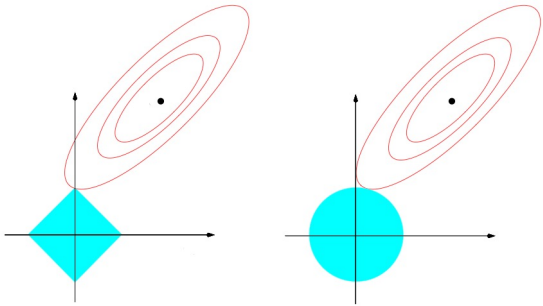
A high degree demand model can shrink implausible oscillating coefficients.

Model statement

Add a coefficient penalty when many correlated or flexible features make the fit unstable.

Common mistake

Penalty strength must be selected with validation data after scaling predictors.



Regression metrics encode error cost

- A metric converts a collection of residuals into a comparison aligned with the decision.
- A delivery planner may care about typical delay, while a safety system may care about rare large misses.

Decision check

No single scalar preserves the location, direction, and tail of every error.

Mean absolute error describes a typical absolute miss

$$\mathrm{MAE} = \frac{1}{N}\sum_{i=1}^{N} |y_i - \hat{y}_i|$$

Unit

The result has the same unit as the response.

Reading

A value of 3.2 minutes means an average absolute timing miss of 3.2 minutes.

Mean squared error emphasizes large misses

$$\mathrm{MSE} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2$$

Consequence

Doubling a residual multiplies its contribution by four.

Unit

The metric is measured in squared response units.

先进计算与数字工程研究中心
ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Root mean squared error restores the response unit

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}$$

Relationship

The root mean squared error is always at least the mean absolute error.

Sensitivity

The square is taken before averaging, so large misses still dominate.

Coefficient of determination compares against the mean baseline

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}$$

Reference

$R^2 = 0$ matches the constant mean predictor on the evaluated data.

Scope

It is not an error measured in response units.

A test coefficient of determination can be negative

What the model says

On new data, $R^2 < 0$ means the model is worse than the evaluated mean baseline under squared error.

What it does not say

Negative does not mean the calculation is impossible; it is a useful warning about generalization.

Use it here

A seasonal demand model trained in summer may underperform the winter mean.

Adjusted coefficient of determination penalizes extra columns

Core idea

The adjusted form accounts for sample size and number of fitted predictors.

Concrete case

Adding an irrelevant feature can increase training coefficient of determination but decrease its adjusted form.

Watch for

It is not a replacement for held out evaluation.

Percentage error fails near zero

$$\text{MAPE} = \frac{100}{N} \sum_i \left| \frac{y_i - \hat{y}_i}{y_i} \right|$$

Appeal

A percentage can compare products with different scales.

Failure

Zero or tiny actual values make the ratio undefined or explosive.

Median absolute error represents the middle case

- Take the median of absolute residuals to reduce the influence of extreme misses.
- If half of delivery predictions miss by at most four minutes, the median absolute error is four minutes.

Decision check

A small median can coexist with unacceptable tail failures.

A baseline gives every metric a reference

What the model says

Compare against a simple constant, seasonal, or last value predictor before celebrating.

What it does not say

A low looking error has no meaning without scale and a reference task.

Use it here

Tomorrow's demand can be compared with today's demand or the same weekday last week.

Residual direction should be reported separately

Core idea

Mean signed error reveals systematic overprediction or underprediction.

Concrete case

A staffing model that underpredicts every busy hour can have moderate absolute error but harmful bias.

Watch for

A near zero signed mean can hide large errors that cancel.

Subgroup metrics expose hidden averaging

Start from the case

A citywide travel time error can hide severe failure in remote districts.

Model statement

Compute the same metrics by relevant region, product, time window, or protected group.

Common mistake

Searching many groups after seeing test results can itself overfit the evaluation.

东进计算与数字工程研究中心
ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Two models can reverse rank under two metrics

Model	Residuals	Mean absolute error	Root mean squared error
A	2, 2, 2, 8	3.5	4.36
B	4, 4, 4, 4	4.0	4.00

Decision

Model A wins on typical absolute miss; Model B wins when large misses are emphasized.

Code can compare several regression views

```
import numpy as np
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
y = np.array([3., 5., 7., 20.])
p = np.array([2., 6., 8., 14.])
print("mean absolute error", mean_absolute_error(y, p))
print("root mean squared error", mean_squared_error(y, p) ** 0.5)
print("coefficient of determination", r2_score(y, p))
```

Next step

Always inspect the residual array and the underlying cases.

A regression scorecard needs more than one number

Start from the case

For delivery time, use mean absolute error as the headline and a high error quantile for tail risk.

Model statement

Report one primary metric, one complementary metric, a baseline, residual plots, and subgroup or time checks.

Common mistake

Adding many metrics without a declared priority makes model selection ambiguous.

先进计算与数字工程研究中心
ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

A binary event needs a probability, not an unlimited line

- Model $P(y = 1 \mid \mathbf{x})$ inside the interval from zero to one.
- Estimate whether a patient will return within thirty days.

Decision check

Applying ordinary linear regression to zero and one labels can produce impossible probabilities.

The sigmoid maps every score to a valid probability

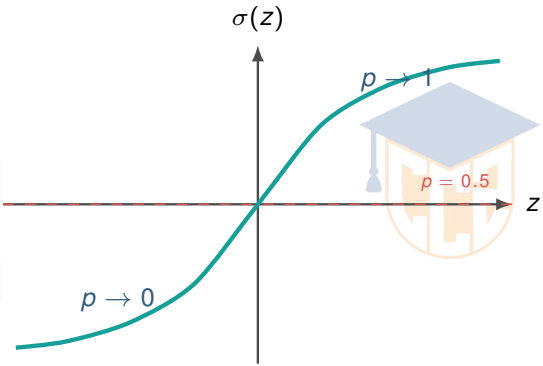
$$\sigma(z) = \frac{1}{1 + e^{-z}}, \qquad p(y = 1 \mid \mathbf{x}) = \sigma(\mathbf{w}^\top \mathbf{x} + b)$$

Range

$0 < \sigma(z) < 1$ for every real score.

Midpoint

$z = 0$ gives probability 0.5.



Log odds restore a linear equation

$$\log \frac{p}{1-p} = \mathbf{w}^\top \mathbf{x} + b$$

Odds

$p/(1-p)$ compares event probability with non-event probability.

Coefficient

A one unit increase in x_j multiplies the odds by e^{w_j} when other variables stay fixed.

The same score change gives different probability changes

Start from the case

Moving a score from zero to one changes probability more than moving it from five to six.

Model statement

Sigmoid sensitivity is greatest near probability one half and smaller near zero or one.

Common mistake

A constant log odds effect is not a constant probability effect.

A Bernoulli model describes each binary outcome

- For $y_i \in \{0, 1\}$, use $p_i^{y_i}(1 - p_i)^{1-y_i}$.
- A returned patient contributes the predicted event probability; a non-returned patient contributes one minus that probability.



Decision check

The model assumes a conditional event probability, not that labels are deterministic.

Likelihood rewards probability assigned to the observed label

$$\mathcal{L}(\mathbf{w}, b) = \prod_i p_i^{y_i} (1 - p_i)^{1 - y_i}$$

Fit

Choose parameters that make the observed labels collectively probable.

Numerics

Use the log likelihood rather than multiplying many small probabilities directly.

Cross entropy is the negative log likelihood

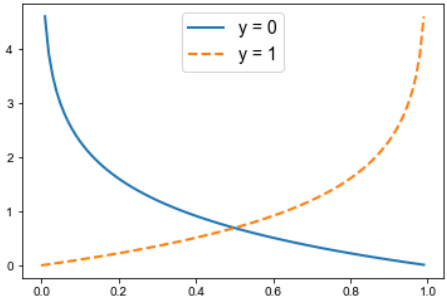
$$-\sum_i [y_i \log p_i + (1 - y_i) \log(1 - p_i)]$$

Correct and confident

A true positive with probability 0.99 receives a very small loss.

Wrong and confident

A true positive with probability 0.01 receives a very large loss.



The logistic objective is convex

Start from the case

Gradient methods can reach a global optimum when a finite optimum exists.

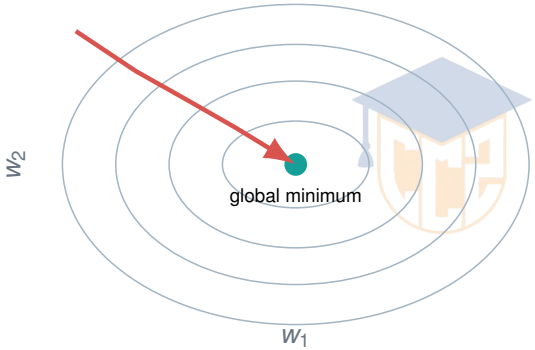
Model statement

For a linear score, binary cross entropy is convex in the coefficients.

Common mistake

Convexity does not protect against bad features, data leakage, or distribution shift.

Convexity removes bad local minima, not data problems.



The gradient is predicted probability minus label

$$\nabla_{\mathbf{w}} J = \mathbf{X}^{\top}(\mathbf{p} - \mathbf{y})$$

Sample signal

A positive sample contributes roughly $p_i - 1$; a negative sample contributes p_i.

Intuition

A missed positive pushes the score upward along its feature direction.

A short program fits and reads probabilities

```
import numpy as np
from sklearn.linear_model import LogisticRegression
X = np.array([[1.0], [2.0], [3.0], [4.0], [5.0], [6.0]])
y = np.array([0, 0, 0, 1, 1, 1])
model = LogisticRegression().fit(X, y)
print(np.round(model.predict_proba([[2.5], [4.5]])[:, 1], 3))
print(model.predict([[2.5], [4.5]]))
```

Read both

Probability and class label answer different questions.

A linear score creates a linear decision boundary

Core idea

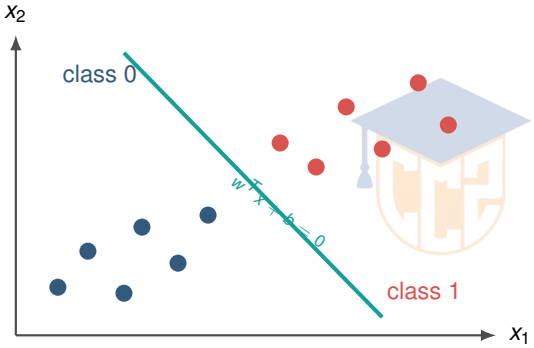
The boundary at probability one half satisfies $\mathbf{w}^\top \mathbf{x} + b = 0$.

Concrete case

Two examination measurements create a straight separating line in a two dimensional plot.

Watch for

A curved class pattern needs transformed features or a different model.



Changing the threshold moves the decision, not the probability model

Start from the case

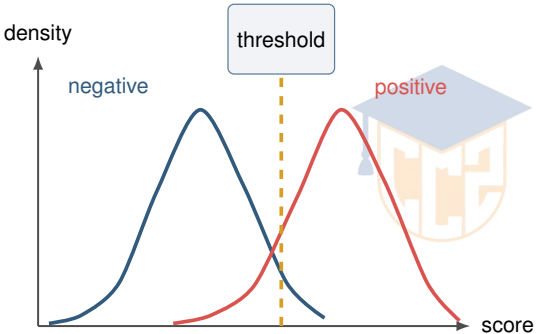
Lowering a fraud alert threshold catches more fraud and also sends more normal payments to review.

Model statement

Predict positive when the estimated probability exceeds a chosen threshold.

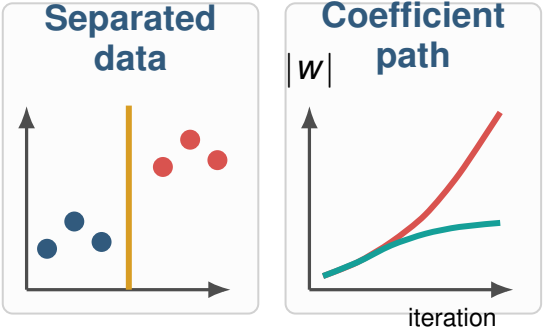
Common mistake

The default threshold of one half has no universal optimality.



Changing the threshold changes actions, not fitted probabilities.

Complete separation can send coefficients toward infinity



Red: no penalty. Teal: with regularization.

- If one line perfectly separates the training classes, unpenalized likelihood may improve as coefficient magnitude grows without bound.
- A tiny classroom dataset has positives only above a score of five and negatives only below.

Decision check

Huge coefficients create extreme probabilities and unstable extrapolation.

Regularization gives separation a finite solution

What the model says

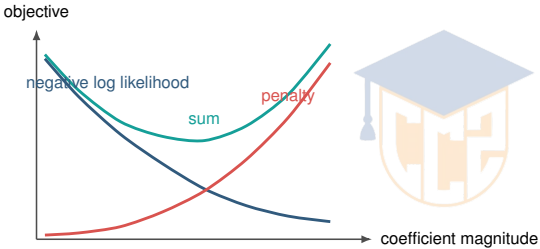
A squared Euclidean norm penalty limits coefficient magnitude and usually yields a stable finite optimum.

What it does not say

Penalty strength affects probability estimates and must be selected within validation.

Use it here

Separated medical training data can still return moderated probabilities.



Coefficient scaling matters before penalized logistic regression

Core idea

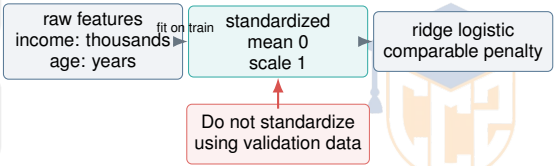
Standardize continuous predictors so the penalty compares coefficients on compatible scales.

Concrete case

Age in years and annual income in currency units otherwise receive uneven shrinkage.

Watch for

Binary indicator coding may be handled differently but must remain consistent.



A penalty acts on coefficient scale, so feature scale matters.

先进计算与数字工程研究中心
ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Odds ratios need a reference and a unit

Start from the case

If a standardized biomarker coefficient is 0.69, a one standard deviation increase roughly doubles the odds.

Model statement

For coefficient w_j, the odds ratio for one unit is e^{w_j}.

Common mistake

An odds ratio of two is not the same as doubling probability.

One versus rest extends logistic regression to many classes

What the model says

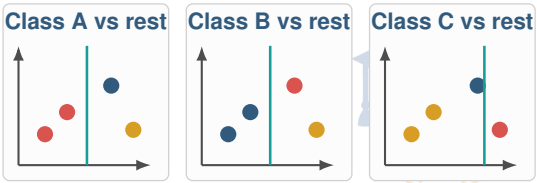
Fit one binary classifier per class against all remaining classes.

What it does not say

Independent binary probabilities may not sum exactly to one without an additional normalization rule.

Use it here

For three flower species, build three class specific scores and choose the largest.



Train one binary score per class, then compare scores.

Softmax models class probabilities jointly

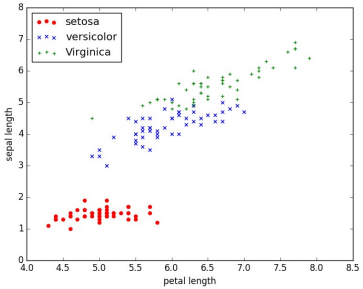
$$P(y = k \mid \mathbf{x}) = \frac{\exp(\mathbf{w}_k^\top \mathbf{x} + b_k)}{\sum_j \exp(\mathbf{w}_j^\top \mathbf{x} + b_j)}$$

Joint constraint

All class probabilities are positive and sum to one.

Reference

One class or an equivalent constraint is needed for identifiable parameters.

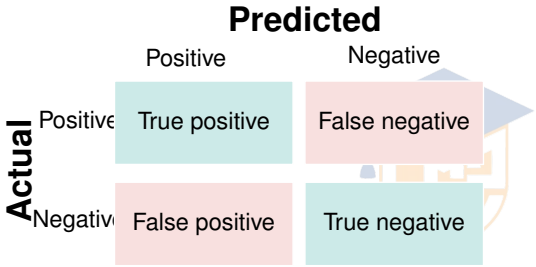


Binary classification outcomes

- After choosing a threshold, every example is positive or negative in both prediction and truth.
- A screening system can detect disease, miss disease, raise a false alarm, or correctly reassure.

Decision check

Collapsing these four outcomes into one accuracy number hides which mistake occurred.



Precision reads a prediction column; recall reads an actual row.

The confusion matrix is a count table with a declared orientation

	Predicted positive	Predicted negative
Actually positive	True positive	False negative
Actually negative	False positive	True negative

First check

Confirm which axis is truth and which axis is prediction before reading any cell.

先进计算与数字工程研究院
ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Accuracy asks how often the final label is correct

$$\text{Accuracy} = \frac{\text{true positives} + \text{true negatives}}{\text{all examples}}$$

Useful

Classes are balanced and both error types have similar cost.

Danger

A dominant majority class can make accuracy look excellent.

先进计算与数字工程研究中心
ADVANCED COMPUTING AND DIGITAL ENGINEERING RESEARCH CENTER

Precision asks whether positive alerts are credible

$$\text{Precision} = \frac{\text{true positives}}{\text{predicted positives}}$$

Viewpoint

Start from all alerts and ask what fraction are real.

Dependence

Precision changes with event prevalence and threshold.

Recall asks how many real positives were found

$$\text{Recall} = \frac{\text{true positives}}{\text{actual positives}}$$

Viewpoint

Start from all real events and ask what fraction were detected.

Tradeoff

Lowering the threshold often raises recall and also false alarms.

Specificity asks how many real negatives were cleared

$$\text{Specificity} = \frac{\text{true negatives}}{\text{actual negatives}}$$

Viewpoint

Start from all real negatives and ask what fraction were correctly rejected.

Complement

One minus specificity is the false positive rate.

The F one score balances precision and recall

$$F_1 = 2\frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

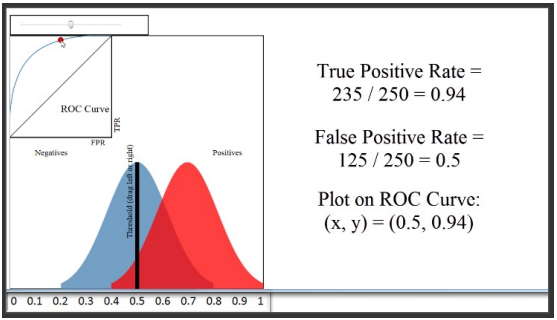
Property

The harmonic mean is pulled down when either component is small.

Assumption

It gives precision and recall symmetric importance and ignores true negatives.

Lower thresholds exchange misses for false alarms



True Positive Rate =
235 / 250 = 0.94

False Positive Rate =
125 / 250 = 0.5

Plot on ROC Curve:
(x, y) = (0.5, 0.94)

Start from the case

A hospital may catch more high risk patients while increasing follow up calls.

Model statement

As the threshold falls, more examples become predicted positive.

Common mistake

The direction is predictable, but the best operating point is task specific.

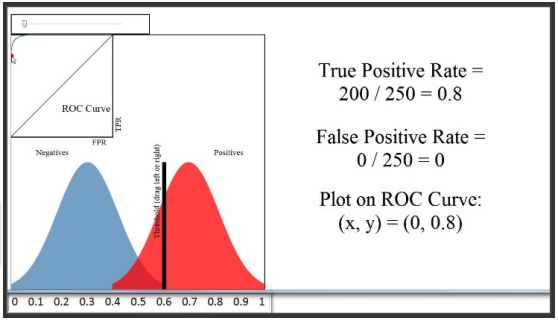
ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Threshold choice is an operating policy

- Select a threshold on validation data using cost, capacity, or a required minimum metric.
- A review team can inspect only two hundred transactions per day.

Decision check

Choosing the threshold on the final test set turns the test into a tuning set.



The receiver operating characteristic curve scans thresholds

What the model says

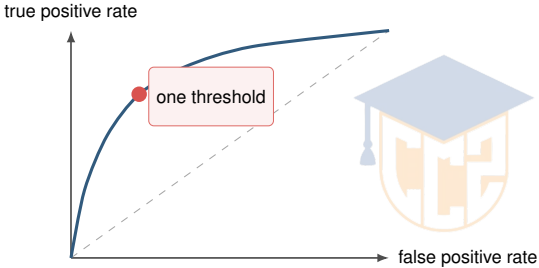
Plot true positive rate against false positive rate while the threshold moves.

What it does not say

The curve does not include prevalence, monetary cost, or review capacity.

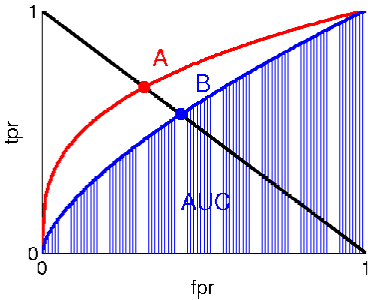
Use it here

Each point is one possible sensitivity and false alarm tradeoff.



Each point is a different operating threshold.

Area under the receiver curve measures ranking



Core idea

It equals the probability that a randomly chosen positive receives a higher score than a randomly chosen negative.

Concrete case

A value near one indicates strong pairwise ranking across thresholds.

Watch for

It does not guarantee calibrated probabilities or good performance at the required operating point.

The precision recall curve focuses on positive alerts

Start from the case

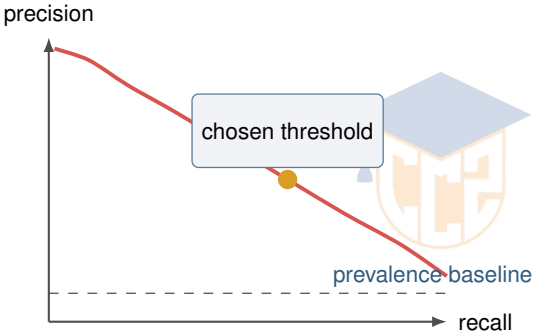
For rare defects, it shows how alert credibility declines while detection coverage rises.

Model statement

Plot precision against recall as the threshold changes.

Common mistake

Its visual baseline depends on positive prevalence.



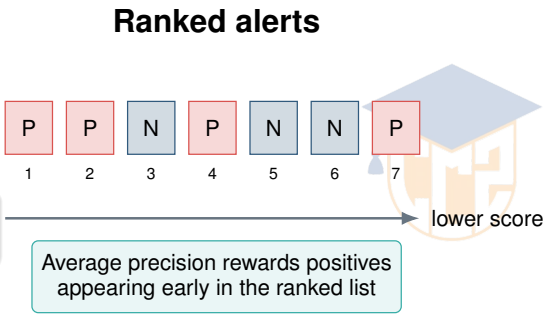
先进计算与数字工程研究中心
ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Average precision summarizes a ranked list

- Weight precision values by the increases in recall along the precision recall curve.
- A model that keeps early alerts clean receives a strong summary.

Decision check

Different interpolation conventions can give slightly different numerical summaries.



A numeric threshold table makes the curve concrete

Threshold	True positives	False positives	Precision	Recall
0.80	18	2	0.90	0.45
0.50	30	10	0.75	0.75
0.20	38	38	0.50	0.95

A multiclass confusion matrix reveals class substitutions

Core idea

Rows and columns show which true class is confused with which predicted class.

Concrete case

A flower classifier may confuse two visually similar species while identifying the third perfectly.

Watch for

Overall accuracy cannot show the direction of these substitutions.

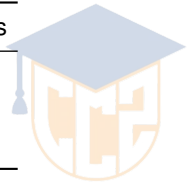
Columns: predicted class

	Class A	Class B	Class C
Actual A	84	9	7
Actual B	6	76	18
Actual C	4	12	84

Off-diagonal cells name the substitutions that need investigation.

Macro and micro averaging answer different questions

Average	How it is formed	Whose voice dominates
Macro	Compute each class, then average	Every class equally
Micro	Pool all class decisions first	Frequent examples
Weighted	Average by class support	Large classes more



Evaluate predicted probabilities and classification errors

```
import numpy as np
from sklearn.metrics import confusion_matrix, precision_score, recall_score
y = np.array([1, 1, 1, 0, 0, 0, 0, 0])
p = np.array([.9, .7, .4, .8, .3, .2, .1, .05])
pred = (p >= .5).astype(int)
print(confusion_matrix(y, pred))
print("precision", precision_score(y, pred))
print("recall", recall_score(y, pred))
```

Audit

Print the threshold and class prevalence beside every scorecard.

Metric selection begins with the costly mistake

What the model says

Name the positive class, identify both error costs, state prevalence, and declare capacity before selecting metrics.

What it does not say

Using the same favorite metric across tasks silently imports the wrong values.

Use it here

Cancer screening prioritizes missed disease differently from promotional email targeting.

Rare positives change what a good score means

Core idea

Class imbalance occurs when one outcome is much less frequent than the other.

Concrete case

Only two transactions in one thousand may be fraudulent.

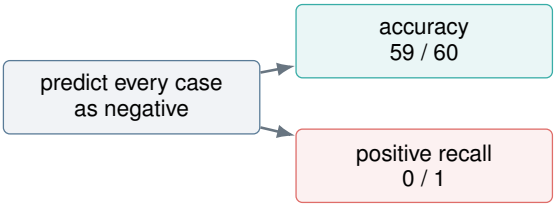
Watch for

Rarity does not by itself make the minority class noisy or unimportant.



A rare event can be hidden inside a sea of correct negatives.

Accuracy under severe class imbalance



High accuracy can coexist with complete failure on the class of interest.

Start from the case

A defect rate of one percent gives ninety nine percent accuracy to an always normal classifier.

Model statement

Predicting every example as the majority class can achieve high accuracy while detecting no positives.

Common mistake

A strong baseline for imbalance must include minority detection, not accuracy alone.

Prevalence changes precision even when rates stay fixed

Population	Prevalence	Recall	False positive rate	Approximate precision
Balanced study	50 percent	90 percent	10 percent	90 percent
Deployment	1 percent	90 percent	10 percent	8 percent

先进计算与数字工程研究中心
ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Stratification preserves class counts but not every dependency

What the model says

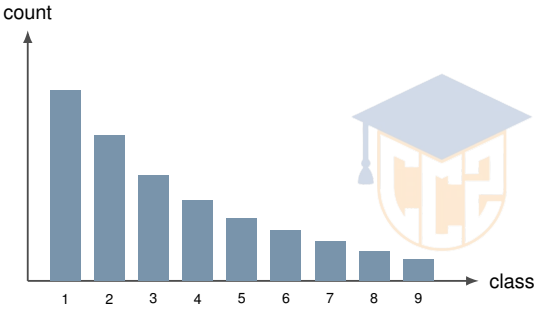
A stratified split keeps approximate class proportions in each partition.

What it does not say

Stratification does not prevent the same patient, machine, or future time from leaking across splits.

Use it here

Rare defects should appear in training, validation, and test sets when sample size allows.



Stratification preserves these counts; it does not preserve groups or time.

Class weights change the training objective

Core idea

Give different loss weights to positive and negative examples.

Concrete case

A missed failure can receive twenty times the loss of a false alarm.

Watch for

A weight changes fitted parameters; it is not the same operation as changing the threshold afterward.

先进计算与数字工程研究中心
ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Threshold tuning uses the original validation distribution

Start from the case

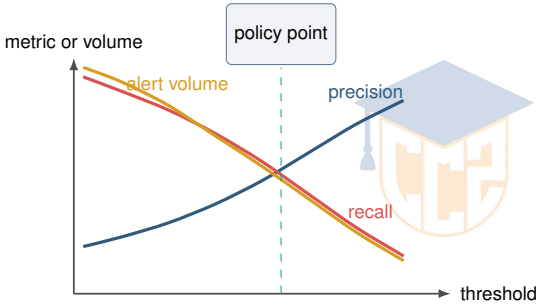
Require at least ninety five percent recall, then select the highest precision feasible point.

Model statement

Choose an operating threshold on untouched validation predictions under the expected deployment prevalence.

Common mistake

Resampling the validation set can distort precision and workload estimates.



Random oversampling repeats minority evidence

- Duplicate or reweight minority examples so they influence fitting more strongly.
- A rare defect image may appear several times in a training epoch.

Decision check

Duplicates do not create new independent information and can encourage memorization.

Random undersampling trades information for balance

What the model says

Discard some majority examples to reduce computation or class dominance.

What it does not say

Aggressive removal can erase important majority subgroups and boundary cases.

Use it here

Keep a manageable subset of normal network traffic while preserving diverse normal patterns.

Synthetic minority interpolation creates local samples

Core idea

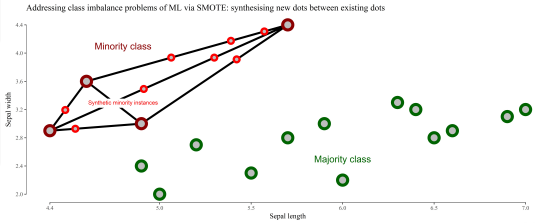
Choose nearby minority examples and interpolate between them to create training points.

Concrete case

Two similar minority transactions define a segment containing a synthetic example.

Watch for

Interpolation assumes the path between neighbors remains a plausible minority region.



Synthetic clouds can cross the wrong boundary

Start from the case

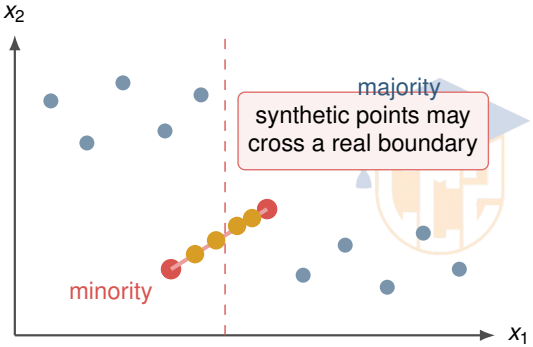
A minority cluster beside a majority cluster may generate ambiguous interpolations.

Model statement

Visualize original and synthetic points together with class neighborhoods.

Common mistake

Categorical variables and mixed scales need specialized handling before interpolation.



Resampling before splitting creates duplicate leakage

- Split first, then fit the resampling step only inside each training fold.
- A duplicated minority record placed in both training and validation makes validation look easy.

Decision check

The final test set must never be resampled using training neighbors.

The precision recall baseline equals prevalence

What the model says

A random ranking has expected precision near the positive fraction.

What it does not say

Curves from datasets with different prevalence are not directly comparable without context.

Use it here

At one percent prevalence, a precision of ten percent is ten times the random baseline.

A cost matrix converts counts into consequences

Outcome	Count	Cost per case	Total cost
False negative	8	5000	40000
False positive	120	20	2400
True positive	42	0	0
True negative	9830	0	0

Lesson

A small count can dominate total cost when its consequence is severe.

Code compares weighting with a fixed baseline

```
import numpy as np
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import precision_score, recall_score
rng = np.random.default_rng(8)
X = rng.normal(size=(500, 2))
y = ((X[:, 0] + X[:, 1] > 2.4)).astype(int)
for weight in [None, "balanced"]:
    model = LogisticRegression(class_weight=weight).fit(X, y)
    pred = model.predict(X)
    print(weight, round(precision_score(y, pred), 3),
          round(recall_score(y, pred), 3))
```

Protocol

In a real study, compare on untouched validation data, not on the training data printed here.

Evaluation protocol for imbalanced data

What the model says

Lock a representative test set, split by groups or time, fit resampling only inside training, choose thresholds on validation, and report deployment oriented metrics.

What it does not say

No balancing technique removes the need for monitoring after deployment.

Use it here

A fraud system reports alert volume, precision, recall, cost, and calibration by week.

先进计算与数字工程研究中心
ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Regression and classification workflows

Stage	Sales amount	Churn event
Output	Continuous value	Probability
Training loss	Squared residual	Cross entropy
Primary check	Absolute error	Precision and recall
Decision layer	Budget forecast	Contact threshold

Check the output, the loss, and the evaluation together

- Regression: a number, a stated unit, and an error metric.
- Classification: probabilities, a threshold, and class-specific errors.
- Fit preprocessing and tuning inside training/validation data.
- Compare with a simple baseline on the final held-out set.

Linear model exam case

华南计算与数字工程研究中心
ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Evidence

A rare defect model has 99.4 percent accuracy, 18 percent precision, 92 percent recall, and rising alert volume.

Task

Diagnose the scorecard, choose two additional checks, and propose a threshold or data action.

Strong answer

Link every action to prevalence, cost, capacity, or leakage evidence.

Linear model self-check

先进计算与数字工程研究中心
ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

1. Explain one coefficient with its unit and reference condition.
2. Choose two regression metrics for a stated error cost.
3. Separate logistic probability from the decision threshold.
4. Read precision and recall from a confusion matrix.
5. Place resampling inside a leakage safe evaluation pipeline.